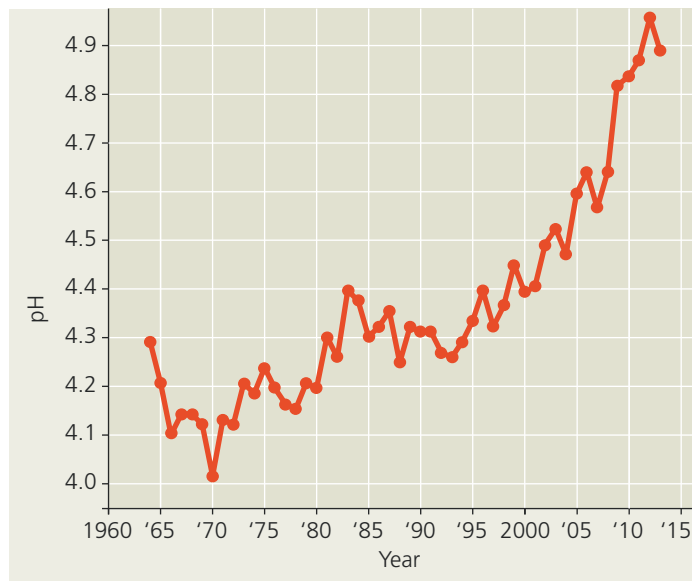


▼ **Figure 56.10** Changes in the pH of precipitation at the Hubbard Brook Experimental Forest, New Hampshire.



MAKE CONNECTIONS Describe the relationship between pH and acidity. (See Concept 3.3.) Overall, is the precipitation in this forest becoming more acidic or less acidic?

because of pollution generated in Great Britain and central Europe. By 1980, the pH of precipitation in large areas of North America and Europe averaged 4.0–4.5 and sometimes dropped as low as 3.0. (To review pH, see Concept 3.3.)

Environmental regulations and new technologies have enabled many countries to reduce sulfur dioxide emissions in recent decades. In the United States, sulfur dioxide emissions decreased more than 75% between 1990 and 2013, gradually reducing the acidity of precipitation (**Figure 56.10**). However, ecologists estimate that it will take decades for aquatic ecosystems to recover. Meanwhile, emissions of nitrogen oxides are increasing in the United States, and emissions of sulfur dioxide and acid precipitation continue to damage forests in central and eastern Europe.

We'll explore the importance of global change for Earth's biodiversity in more detail in Concept 56.4, where we examine such factors as climate change and ozone depletion.

Can Extinct Species Be Resurrected?

To the best of our knowledge, extinction has always been forever. Some scientists are nevertheless trying to use cloning techniques (see Concept 20.3) to resurrect species that have become extinct. For example, researchers used this approach in an attempt to reverse the extinction of the Pyrenean ibex, *Capra pyrenaica pyrenaica*. In 1999, researchers removed a skin sample from the last living individual, a female, and froze it. When that individual died a year later, scientists used the frozen cells to try to produce more ibex. Eventually, one ibex was born in 2009. Sadly, it lived for only 7 minutes before succumbing to lung defects similar to those observed in other cloned animals.

More recently, researchers completed an early step toward the possible use of cloning to prevent the extinction of the northern white rhinoceros (*Ceratotherium simum cottoni*). The last *C. simum cottoni* male died in 2018—dooming the northern white rhinoceros to eventual extinction. That same year, researchers successfully created hybrid embryos using sperm from northern white rhinoceroses and eggs from closely related southern white rhinoceroses, *C. simum simum* (the few living females from the northern subspecies all had severe reproductive issues).

Many steps remain before the northern white rhinoceros could be considered resurrected from extinction. Moreover, even if scientists eventually can resurrect one or a few species, that approach will not solve the extinction crisis we face today. To do that, we must take steps to clean up the environment and protect natural habitat for the thousands of species currently threatened by extinction.

CONCEPT CHECK 56.1

1. Explain why it is too narrow to define the biodiversity crisis as simply a loss of species.
2. Identify the four main threats to biodiversity and explain how each damages diversity.
3. **WHAT IF?** Imagine two populations of a fish species, one in the Mediterranean Sea and one in the Caribbean Sea. Now imagine two scenarios: (1) The populations breed separately, and (2) adults of both populations migrate yearly to the North Atlantic to interbreed. Which scenario would result in a greater loss of genetic diversity if the Mediterranean population were harvested to extinction? Explain your answer.

For suggested answers, see Appendix A.

CONCEPT 56.2

Population conservation focuses on population size, genetic diversity, and critical habitat

Biologists who work on conservation at the population and species levels use two main approaches. One approach focuses on populations that are small and hence often vulnerable, while the other emphasizes critical habitat.

Extinction Risks in Small Populations

Small populations are particularly vulnerable to overharvesting, habitat loss, and the other threats to biodiversity that you read about in Concept 56.1. After such factors have reduced a population's size to a small number of individuals, the small size itself can push the population to extinction.

The Extinction Vortex: Evolutionary Implications of Small Population Size

EVOLUTION A small population is vulnerable to inbreeding and genetic drift, which can draw the population down an **extinction vortex** toward smaller and smaller population